

Holden Hess

919-710-7276 | holdenhess.work@gmail.com | [GitHub](#) | [LinkedIn](#)

1 PROFESSIONAL SUMMARY

I graduated from Appalachian State University with hands-on experience in C/C++, Python, and SQL. I am skilled in software development, problem-solving, and debugging with a strong foundation in functional and object-oriented programming. I am looking for an entry level position to get my foot in the door of the industry.

2 EDUCATION

BACHELOR OF SCIENCE IN COMPUTER SCIENCE, CUM LAUDE

- Appalachian State University | Graduated: Dec 2024
- Relevant Coursework: Data Structures, Programming Languages, Computer Systems, Neural Networks, Linear Algebra, Software Engineering, Systems Administration and Cybersecurity

3 SKILLS

- ❖ **Languages:** C/C++, Java, Python, Haskell, HTML/CSS/JavaScript, Kotlin, x86/y86 Assembly
- ❖ **Frameworks & Tools:** Git, Android Studio, Visual Studio, Linux, Maven, Javalin
- ❖ **Databases:** MySQL, JDBC
- ❖ **Other Skills:** Agile/Waterfall Development, UML, Problem Solving, Debugging, Functional & Object-Oriented Programming

4 PROJECTS

SIMULATED Y86 ARCHITECTURE IN C++ (SPRING 2024)

- Developed a simulation of the Y86 architecture using C++ to gain in-depth understanding of CPU pipeline processes and assembly language.
- Implemented core elements of instruction cycles and handling of pipeline hazards, improving understanding of low-level system operations.

TEAM SQL PROJECT – AUTONOMOUS CARS DATABASE (FALL 2023)

- Collaborated with a team to design and create a MySQL database to store information related to autonomous cars.
- Gained practical knowledge in database structuring, normalization, and querying, while learning the fundamentals of SQL in a real-world context.

ANDROID APP FOR HUNGER AND HEALTH COALITION (FALL 2024)

- Worked with a team of three to design and implement the front end of an Android application for a local nonprofit in Boone, NC.
- Developed user-friendly interfaces in Android Studio and integrated the app with a backend database to display live events to app users.
- Enhanced teamwork and project management skills while delivering a tailored solution to the Hunger and Health Coalition.

NEURAL NETWORK FROM SCRATCH (FALL 2024)

- Developed a deep learning framework in python inspired by TensorFlow, supporting arbitrary input shapes, user-defined depth, multiple loss and accuracy functions, and L1/L2 regularization using only NumPy package.
- Implemented forward and backward propagation, activation functions, and gradient descent optimizations without external deep learning libraries.
- Designed a flexible architecture allowing users to construct and train custom neural networks with dynamic layer configurations.